

Newman-Penrose Formalism Calculations

Stephani et al., Equation (33.44a)

Metric

$$\begin{aligned}
 ds^2 = & -dt \otimes dt + \left(\frac{\Phi^2(t) (k^2 A^2 + 4kAU + 4U^2) x^4}{k^2 x^4 + k^2 y^4 + 2(k^2 x^2 + 2k) y^2 + 4kx^2 + 4} + \frac{\Phi^2(t) (k^2 A^2 + 4kAU + 4U^2) y^4}{k^2 x^4 + k^2 y^4 + 2(k^2 x^2 + 2k) y^2 + 4kx^2 + 4} + \frac{8\Phi^2(t) (kAV_r + 2UV_r) x^3}{k^2 x^4 + k^2 y^4 + 2(k^2 x^2 + 2k) y^2 + 4kx^2 + 4} \right. \\
 & - \frac{8\Phi^2(t) (kAV_i + 2UV_i) y^3}{k^2 x^4 + k^2 y^4 + 2(k^2 x^2 + 2k) y^2 + 4kx^2 + 4} + \frac{4\Phi^2(t) (kA^2 + 2AU + 4V_r^2) x^2}{k^2 x^4 + k^2 y^4 + 2(k^2 x^2 + 2k) y^2 + 4kx^2 + 4} \\
 & + \frac{2\Phi^2(t) [(k^2 A^2 + 4kAU + 4U^2) x^2 + 4(kAV_r + 2UV_r) x + 2kA^2 + 4AU + 8V_i^2] y^2}{k^2 x^4 + k^2 y^4 + 2(k^2 x^2 + 2k) y^2 + 4kx^2 + 4} + \frac{16\Phi^2(t) x A V_r}{k^2 x^4 + k^2 y^4 + 2(k^2 x^2 + 2k) y^2 + 4kx^2 + 4} \\
 & \left. - \frac{8\Phi^2(t) [(kAV_i + 2UV_i) x^2 + 4x V_i V_r + 2A V_i] y}{k^2 x^4 + k^2 y^4 + 2(k^2 x^2 + 2k) y^2 + 4kx^2 + 4} + \frac{4\Phi^2(t) A^2}{k^2 x^4 + k^2 y^4 + 2(k^2 x^2 + 2k) y^2 + 4kx^2 + 4} \right) dr \\
 & \otimes dr + \frac{4\Phi^2(t)}{k^2 x^4 + k^2 y^4 + 2(k^2 x^2 + 2k) y^2 + 4kx^2 + 4} dx \otimes dx + \frac{4\Phi^2(t)}{k^2 x^4 + k^2 y^4 + 2(k^2 x^2 + 2k) y^2 + 4kx^2 + 4} dy \otimes dy
 \end{aligned}$$

Coordinates

$$(x^\mu) = (t, r, x, y)$$

Null Tetrad

Covariant components

$$l = \sqrt{\frac{1}{2}} dt + \frac{\Phi(t)}{\sqrt{2} (1 + \frac{(x^2 + y^2) k}{2})} \left(A (1 + \frac{(x^2 + y^2) k}{2}) + (x^2 + y^2) U + 2 V_r x - 2 V_i y \right) dr$$

$$n = \sqrt{\frac{1}{2}} dt - \frac{1}{\Phi(t) \sqrt{2} (1 + \frac{(x^2 + y^2) k}{2})} \left(A (1 + \frac{(x^2 + y^2) k}{2}) + (x^2 + y^2) U + 2 V_r x - 2 V_i y \right) dr$$

$$m = \left(\frac{2 \sqrt{\frac{1}{2}} \Phi(t)}{kx^2 + ky^2 + 2} \right) dx + \left(\frac{2i \sqrt{\frac{1}{2}} \Phi(t)}{kx^2 + ky^2 + 2} \right) dy$$

$$\bar{m} = \left(\frac{2\sqrt{\frac{1}{2}}\Phi(t)}{kx^2 + ky^2 + 2} \right) dx + \left(-\frac{2i\sqrt{\frac{1}{2}}\Phi(t)}{kx^2 + ky^2 + 2} \right) dy$$

Contravariant components

$$l = \left(-\frac{1}{2}\sqrt{2} \right) dt + \left(\frac{\Phi(t)\sqrt{2}[(kA(t, r) + 2U(r))x^2 + (kA(t, r) + 2U(r))y^2 - 4yV_i(r) + 4xV_r(r) + 2A(t, r)]|kx^2 + ky^2 + 2|}{2(kx^2A(t, r) + ky^2A(t, r) + 2x^2U(r) + 2y^2U(r) - 4yV_i(r) + 4xV_r(r) + 2A(t, r))^2} \right) dr$$

$$n = \left(-\frac{1}{2}\sqrt{2} \right) dt + \left(-\frac{\Phi(t)\sqrt{2}[(kA(t, r) + 2U(r))x^2 + (kA(t, r) + 2U(r))y^2 - 4yV_i(r) + 4xV_r(r) + 2A(t, r)]|kx^2 + ky^2 + 2|}{2(kx^2A(t, r) + ky^2A(t, r) + 2x^2U(r) + 2y^2U(r) - 4yV_i(r) + 4xV_r(r) + 2A(t, r))^2} \right) dr$$

$$m = \left(\frac{\sqrt{2}kx^2}{4\Phi(t)} + \frac{\sqrt{2}ky^2}{4\Phi(t)} + \frac{\sqrt{2}}{2\Phi(t)} \right) dx + \left(\frac{i\sqrt{2}kx^2}{4\Phi(t)} + \frac{i\sqrt{2}ky^2}{4\Phi(t)} + \frac{i\sqrt{2}}{2\Phi(t)} \right) dy$$

$$\bar{m} = \left(\frac{\sqrt{2}kx^2}{4\Phi(t)} + \frac{\sqrt{2}ky^2}{4\Phi(t)} + \frac{\sqrt{2}}{2\Phi(t)} \right) dx + \left(-\frac{i\sqrt{2}kx^2}{4\Phi(t)} - \frac{i\sqrt{2}ky^2}{4\Phi(t)} - \frac{i\sqrt{2}}{2\Phi(t)} \right) dy$$

Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}\partial_t + \frac{\Phi(t)\sqrt{2}[(kA(t, r) + 2U(r))x^2 + (kA(t, r) + 2U(r))y^2 - 4yV_i(r) + 4xV_r(r) + 2A(t, r)]|kx^2 + ky^2 + 2|}{2(kx^2A(t, r) + ky^2A(t, r) + 2x^2U(r) + 2y^2U(r) - 4yV_i(r) + 4xV_r(r) + 2A(t, r))^2} \partial_r$$

$$\Delta = -\frac{1}{2}\sqrt{2}\partial_t - \frac{\Phi(t)\sqrt{2}[(kA(t, r) + 2U(r))x^2 + (kA(t, r) + 2U(r))y^2 - 4yV_i(r) + 4xV_r(r) + 2A(t, r)]|kx^2 + ky^2 + 2|}{2(kx^2A(t, r) + ky^2A(t, r) + 2x^2U(r) + 2y^2U(r) - 4yV_i(r) + 4xV_r(r) + 2A(t, r))^2} \partial_r$$

$$\delta = \frac{\sqrt{2}kx^2}{4\Phi(t)} + \frac{\sqrt{2}ky^2}{4\Phi(t)} + \frac{\sqrt{2}}{2\Phi(t)} \partial_x + \frac{i\sqrt{2}kx^2}{4\Phi(t)} + \frac{i\sqrt{2}ky^2}{4\Phi(t)} + \frac{i\sqrt{2}}{2\Phi(t)} \partial_y$$

$$\bar{\delta} = \frac{\sqrt{2}kx^2}{4\Phi(t)} + \frac{\sqrt{2}ky^2}{4\Phi(t)} + \frac{\sqrt{2}}{2\Phi(t)} \partial_x - \frac{i\sqrt{2}kx^2}{4\Phi(t)} - \frac{i\sqrt{2}ky^2}{4\Phi(t)} - \frac{i\sqrt{2}}{2\Phi(t)} \partial_y$$

Spin Coefficients

$$\rho = \frac{\sqrt{2} \frac{\partial}{\partial t} \Phi(t)}{2 \Phi(t)}$$

$$\sigma = 0$$

$$\begin{aligned} \kappa = & -\frac{i \sqrt{2} k x^2 V_i(r)}{2 (k x^2 A(t, r) \Phi(t) + k y^2 A(t, r) \Phi(t) + 2 x^2 \Phi(t) U(r) + 2 y^2 \Phi(t) U(r) - 4 y \Phi(t) V_i(r) + 4 x \Phi(t) V_r(r) + 2 A(t, r) \Phi(t))} \\ & + \frac{\sqrt{2} k x y V_i(r)}{k x^2 A(t, r) \Phi(t) + k y^2 A(t, r) \Phi(t) + 2 x^2 \Phi(t) U(r) + 2 y^2 \Phi(t) U(r) - 4 y \Phi(t) V_i(r) + 4 x \Phi(t) V_r(r) + 2 A(t, r) \Phi(t)} \\ & + \frac{i \sqrt{2} k y^2 V_i(r)}{2 (k x^2 A(t, r) \Phi(t) + k y^2 A(t, r) \Phi(t) + 2 x^2 \Phi(t) U(r) + 2 y^2 \Phi(t) U(r) - 4 y \Phi(t) V_i(r) + 4 x \Phi(t) V_r(r) + 2 A(t, r) \Phi(t))} \\ & - \frac{\sqrt{2} k x^2 V_r(r)}{2 (k x^2 A(t, r) \Phi(t) + k y^2 A(t, r) \Phi(t) + 2 x^2 \Phi(t) U(r) + 2 y^2 \Phi(t) U(r) - 4 y \Phi(t) V_i(r) + 4 x \Phi(t) V_r(r) + 2 A(t, r) \Phi(t))} \\ & - \frac{i \sqrt{2} k x y V_r(r)}{k x^2 A(t, r) \Phi(t) + k y^2 A(t, r) \Phi(t) + 2 x^2 \Phi(t) U(r) + 2 y^2 \Phi(t) U(r) - 4 y \Phi(t) V_i(r) + 4 x \Phi(t) V_r(r) + 2 A(t, r) \Phi(t)} \\ & + \frac{\sqrt{2} k y^2 V_r(r)}{2 (k x^2 A(t, r) \Phi(t) + k y^2 A(t, r) \Phi(t) + 2 x^2 \Phi(t) U(r) + 2 y^2 \Phi(t) U(r) - 4 y \Phi(t) V_i(r) + 4 x \Phi(t) V_r(r) + 2 A(t, r) \Phi(t))} \\ & + \frac{\sqrt{2} x U(r)}{k x^2 A(t, r) \Phi(t) + k y^2 A(t, r) \Phi(t) + 2 x^2 \Phi(t) U(r) + 2 y^2 \Phi(t) U(r) - 4 y \Phi(t) V_i(r) + 4 x \Phi(t) V_r(r) + 2 A(t, r) \Phi(t)} \\ & + \frac{i \sqrt{2} y U(r)}{k x^2 A(t, r) \Phi(t) + k y^2 A(t, r) \Phi(t) + 2 x^2 \Phi(t) U(r) + 2 y^2 \Phi(t) U(r) - 4 y \Phi(t) V_i(r) + 4 x \Phi(t) V_r(r) + 2 A(t, r) \Phi(t)} \\ & - \frac{i \sqrt{2} V_i(r)}{k x^2 A(t, r) \Phi(t) + k y^2 A(t, r) \Phi(t) + 2 x^2 \Phi(t) U(r) + 2 y^2 \Phi(t) U(r) - 4 y \Phi(t) V_i(r) + 4 x \Phi(t) V_r(r) + 2 A(t, r) \Phi(t)} \\ & + \frac{\sqrt{2} V_r(r)}{k x^2 A(t, r) \Phi(t) + k y^2 A(t, r) \Phi(t) + 2 x^2 \Phi(t) U(r) + 2 y^2 \Phi(t) U(r) - 4 y \Phi(t) V_i(r) + 4 x \Phi(t) V_r(r) + 2 A(t, r) \Phi(t)} \end{aligned}$$

$$\mu = -\frac{\sqrt{2} \frac{\partial}{\partial t} \Phi(t)}{2 \Phi(t)}$$

$$\lambda = 0$$

$$\begin{aligned}
\nu = & -\frac{i\sqrt{2}kx^2V_i(r)}{2(kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t))} \\
& -\frac{\sqrt{2}kxyV_i(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& +\frac{i\sqrt{2}ky^2V_i(r)}{2(kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t))} \\
& +\frac{\sqrt{2}kx^2V_r(r)}{2(kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t))} \\
& -\frac{i\sqrt{2}kxyV_r(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& -\frac{\sqrt{2}ky^2V_r(r)}{2(kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t))} \\
& -\frac{\sqrt{2}xU(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& +\frac{i\sqrt{2}yU(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& -\frac{i\sqrt{2}V_i(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& -\frac{\sqrt{2}V_r(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)}
\end{aligned}$$

$$\alpha = \frac{\sqrt{2}kx}{4\Phi(t)} - \frac{i\sqrt{2}ky}{4\Phi(t)}$$

$$\beta = -\frac{\sqrt{2}kx}{4\Phi(t)} - \frac{i\sqrt{2}ky}{4\Phi(t)}$$

$$\begin{aligned}
\tau = & \frac{i\sqrt{2}kx^2V_i(r)}{2(kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t))} \\
& - \frac{\sqrt{2}kxyV_i(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& - \frac{i\sqrt{2}ky^2V_i(r)}{2(kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t))} \\
& + \frac{\sqrt{2}kx^2V_r(r)}{2(kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t))} \\
& + \frac{i\sqrt{2}kxyV_r(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& - \frac{\sqrt{2}ky^2V_r(r)}{2(kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t))} \\
& - \frac{\sqrt{2}xU(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& - \frac{i\sqrt{2}yU(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& + \frac{i\sqrt{2}V_i(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& - \frac{\sqrt{2}V_r(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)}
\end{aligned}$$

$$\begin{aligned}
\pi = & \frac{i\sqrt{2}kx^2V_i(r)}{2(kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t))} \\
& + \frac{\sqrt{2}kxyV_i(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& - \frac{i\sqrt{2}ky^2V_i(r)}{2(kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t))} \\
& - \frac{\sqrt{2}kx^2V_r(r)}{2(kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t))} \\
& + \frac{i\sqrt{2}kxyV_r(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& + \frac{\sqrt{2}ky^2V_r(r)}{2(kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t))} \\
& + \frac{\sqrt{2}xU(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& - \frac{i\sqrt{2}yU(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& + \frac{i\sqrt{2}V_i(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)} \\
& + \frac{\sqrt{2}V_r(r)}{kx^2A(t,r)\Phi(t) + ky^2A(t,r)\Phi(t) + 2x^2\Phi(t)U(r) + 2y^2\Phi(t)U(r) - 4y\Phi(t)V_i(r) + 4x\Phi(t)V_r(r) + 2A(t,r)\Phi(t)}
\end{aligned}$$

$$\begin{aligned}
\epsilon = & - \frac{kx^2\Phi(t) \frac{\partial}{\partial t} A(t, r)}{2(\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t))} \\
& - \frac{ky^2\Phi(t) \frac{\partial}{\partial t} A(t, r)}{2(\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t))} \\
& - \frac{kx^2A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2(\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t))} \\
& - \frac{ky^2A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2(\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t))} \\
& - \frac{x^2U(r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t)} \\
& - \frac{y^2U(r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t)} \\
& + \frac{2yV_i(r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t)} \\
& - \frac{2xV_r(r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t)} \\
& - \frac{\Phi(t) \frac{\partial}{\partial t} A(t, r)}{\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t)} \\
& - \frac{A(t, r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t)}
\end{aligned}$$

$$\begin{aligned}
\gamma = & \frac{kx^2\Phi(t) \frac{\partial}{\partial t} A(t, r)}{2(\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t))} \\
& + \frac{ky^2\Phi(t) \frac{\partial}{\partial t} A(t, r)}{2(\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t))} \\
& + \frac{kx^2A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2(\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t))} \\
& + \frac{ky^2A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2(\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t))} \\
& + \frac{x^2U(r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t)} \\
& + \frac{y^2U(r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t)} \\
& - \frac{2yV_i(r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t)} \\
& + \frac{2xV_r(r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t)} \\
& + \frac{\Phi(t) \frac{\partial}{\partial t} A(t, r)}{\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t)} \\
& + \frac{A(t, r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2}kx^2A(t, r)\Phi(t) + \sqrt{2}ky^2A(t, r)\Phi(t) + 2\sqrt{2}x^2\Phi(t)U(r) + 2\sqrt{2}y^2\Phi(t)U(r) - 4\sqrt{2}y\Phi(t)V_i(r) + 4\sqrt{2}x\Phi(t)V_r(r) + 2\sqrt{2}A(t, r)\Phi(t)}
\end{aligned}$$

Ricci Tensor Components

$$\begin{aligned}
\Phi_{00} = & \frac{kx^2\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{ky^2\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{kx^2 A(t, r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{ky^2 A(t, r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{kx^2 A(t, r) \Phi(t) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{ky^2 A(t, r) \Phi(t) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{x^2 U(r) \frac{\partial}{\partial t} \Phi(t)^2}{kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2} \\
& + \frac{y^2 U(r) \frac{\partial}{\partial t} \Phi(t)^2}{kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2} \\
& - \frac{x^2 \Phi(t) U(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2} \\
& - \frac{y^2 \Phi(t) U(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2} + \dots
\end{aligned}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned}
\Phi_{11} = & - \frac{kx^2\Phi(t)^2 \frac{\partial^2}{(\partial t)^2} A(t, r)}{4 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{ky^2\Phi(t)^2 \frac{\partial^2}{(\partial t)^2} A(t, r)}{4 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{kx^2\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{ky^2\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{kx^2 A(t, r) \frac{\partial}{\partial t} \Phi(t)^2}{4 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{ky^2 A(t, r) \frac{\partial}{\partial t} \Phi(t)^2}{4 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{kx^2 A(t, r) \Phi(t) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{4 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{ky^2 A(t, r) \Phi(t) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{4 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{k^2 x^2 A(t, r)}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{k^2 y^2 A(t, r)}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} + \dots
\end{aligned}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\begin{aligned}
\Phi_{22} = & \frac{kx^2\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{ky^2\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{kx^2 A(t, r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{ky^2 A(t, r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{kx^2 A(t, r) \Phi(t) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{ky^2 A(t, r) \Phi(t) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{x^2 U(r) \frac{\partial}{\partial t} \Phi(t)^2}{kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2} \\
& + \frac{y^2 U(r) \frac{\partial}{\partial t} \Phi(t)^2}{kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2} \\
& - \frac{x^2 \Phi(t) U(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2} \\
& - \frac{y^2 \Phi(t) U(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2} + \dots
\end{aligned}$$

$$\begin{aligned}
\Lambda = & \frac{kx^2\Phi(t)^2 \frac{\partial^2}{(\partial t)^2} A(t, r)}{12 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{ky^2\Phi(t)^2 \frac{\partial^2}{(\partial t)^2} A(t, r)}{12 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{kx^2\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{3 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{ky^2\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{3 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{kx^2 A(t, r) \frac{\partial}{\partial t} \Phi(t)^2}{4 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{ky^2 A(t, r) \frac{\partial}{\partial t} \Phi(t)^2}{4 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{kx^2 A(t, r) \Phi(t) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{4 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{ky^2 A(t, r) \Phi(t) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{4 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{k^2 x^2 A(t, r)}{6 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& + \frac{k^2 y^2 A(t, r)}{6 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} + \dots
\end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned}
\Psi_2 = & - \frac{kx^2\Phi(t)^2 \frac{\partial^2}{(\partial t)^2} A(t, r)}{6 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{ky^2\Phi(t)^2 \frac{\partial^2}{(\partial t)^2} A(t, r)}{6 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{kx^2\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{6 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{ky^2\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{6 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{k^2 x^2 A(t, r)}{3 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{k^2 y^2 A(t, r)}{3 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{kx^2 U(r)}{3 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{ky^2 U(r)}{3 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{\Phi(t)^2 \frac{\partial^2}{(\partial t)^2} A(t, r)}{3 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{3 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{2kA(t, r)}{3 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)} \\
& - \frac{2U(r)}{3 \left(kx^2 A(t, r) \Phi(t)^2 + ky^2 A(t, r) \Phi(t)^2 + 2x^2 \Phi(t)^2 U(r) + 2y^2 \Phi(t)^2 U(r) - 4y\Phi(t)^2 V_i(r) + 4x\Phi(t)^2 V_r(r) + 2A(t, r) \Phi(t)^2 \right)}
\end{aligned}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type " D "